



Technical Service Bulletin

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Cylinder Head Repair Procedures

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic is adding service tools and steps to the cylinder head procedure for rebuilding and repairing the cylinder head. There are repair criteria that are unique on the Signature/ISX and QSX15 cylinder head. The repair procedure does **not** include steps for grinding valves or valve seat inserts. If a valve fails to pass inspection, or the valve and seat together will **not** seal properly then the valve and seat will have to be replaced.

Valve and valve seat insert inspection criteria:

Inspect for proper valve recess

Inspect the valve for any cracks, large scratches, and correct stem dimension

Inspect for burnt valves and/or seat combination.

Vacuum test the valve and seat combination.

Repair Guidelines:

If the valve, or valve and seat does not pass inspection, both the valve and the seat need to be replaced.

If a valve and seat are replaced, the appropriate repair is to replace the other valve and seat in the pair.

The new parts from Cummins, Inc. are pre-machined so that there is no need to lap the valve and seat together.

Early generation engines are equipped with a 30 degree intake valve and seat angle. The service parts are of a 20 degree seat angle design. It is appropriate to replace a 30 degree valve and insert with 20 degree valve and insert service parts.

Table 1, Service Parts For The Signature/ISX and QSX15 Cylinder Head

Part Description	Part Number
Exhaust Seat Insert	3679537
Intake Seat Insert	4059215

Table 1, Service Parts For The Signature/ISX and QSX15 Cylinder Head	
Part Description	Part Number
Exhaust Valve	4059327
Intake Valve	4059326
Threaded Plug - Fuel Passage	3679354
Threaded Plug - Coolant Passage	3678611
Valve Spring Guide	3680318
Collet	3680883
Valve Spring Retainer	3680884
Valve Guide	4026657
Valve Stem Seal - Intake	4026660 (guide: 4026657)
Valve Stem Seal - Exhaust	4026791 (guide: 4026657)
Valve Stem Seal - Intake (old style)	4003966 (guide: 3680867*)
Valve Stem Seal - Exhaust (old style)	3606766 (guide: 3680867*)
Injector Cam Bushing	4026423
Expansion Plug	3015865
Expansion Plug	3813247
* Valve guide 3680867 is obsolete, if replacement is required use 4026657	

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